Forest Ecology and Management 453 (2019) 117558

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

The extent to which an unforeseen biotic disturbance can challenge timber expectations

Alex J. Woods[a,*], Martin Watts[b]

[a] BC Ministry of Forests Lands and Natural Resource Operations, Skeena Region, Bag 6000, Smithers, BC V0J 2N0, Canada
[b] FORCOMP Forestry Consulting Ltd., A-827 Queens Avenue, Victoria, BC V8T 4V8, Canada

ARTICLE INFO

Keywords:
Managed forest productivity
Biotic disturbance agents
Forest pathogens
Uncertainty
Interior spruce
Tomentosus root disease

ABSTRACT

Forest pathogens are one of the key drivers of the dynamics and diversity of forested landscapes, contributing to structural and compositional complexity at multiple scales in space and time. Growth and yield models help guide forest management decisions but the roles of abiotic and biotic disturbance agents including pathogens are often overlooked. In our study of mid-rotation spruce plantations in northwest British Columbia (BC), Canada, we examine the influence of tomentosus root disease on the oldest managed stands in a previously unmanaged landscape allowing us to witness pathogen expression in a first rotation setting over the longest duration available. Our 10 one-hectare stem-mapped plots capture the spatial variability of root disease expression and their large size, coupled with their random selection without any prior knowledge of the pathogen, make our study unique. We found tomentosus root disease accounted for nearly 90% of both the number and the volume of spruce trees that died and that the incidence of the disease more than doubled over a 16-year period from stand age ~30–46. The volume growth of trees that were initially healthy but became infected was 17% less than that of the trees that remained healthy, while the volume growth of trees that were infected since plot establishment was 28% less than that of trees that remained healthy. We found a weak relationship between infected stumps from the previous stand and the condition of trees surrounding those stumps at age 30, but more critically we found a similarly weak relationship between infected trees at age 30 and the condition of the neighbouring trees 16 years later. Using the growth and yield model TASS II v2.7.75, we found model projections of spruce volume exceeded measured plot volume by an average of 30.5%. TASS is based on the same foundational tenets that are the dominant paradigm in forestry where the prediction of forest growth is based on the assessment of site index (dominant stand height and age), Eichhorn's rule (which implies that when the dominant tree height is known, stand yield can be easily estimated now and then projected into the future), and the thinning response hypothesis. Our results reflect the influential role of a forest pathogen that is operating on a different scale and following a different set of rules. This, in combination with the unexpected behaviour of the forest pathogen itself, represents a multifaceted challenge in forecasting forest growth.

1. Introduction

The practice of forestry is similar to many natural resource management endeavours where the desire to reach specific objectives often leads to oversimplification and compartmentalized thinking (Holling and Meffe, 1996; Seidl et al., 2008; Puettmann et al., 2009). Forest pathogens are key drivers of the dynamics and diversity of forested landscapes (Holdenrieder et al., 2004), contributing to structural and compositional complexity at multiple scales in space and time (Kosiba et al., 2018). Although considerable effort has been made to model their behaviour and impacts (eg. McNamee et al., 1989; Pukkala et al., 2005; Peet and Hunt, 2005), the influence of forest pathogens is often overlooked in the traditional European forest management model which is built on the premise of regulated, disease and decay free forests (Hansen and Goheen, 2000). This oversight is not confined to traditional forest management. More broadly applied ecosystem models have similarly failed to comprehensively synthesize the effects of biotic disturbance agents including pathogens (Flower and Gonzalez-Meler, 2015), despite long-standing evidence that pathogens can profoundly affect forest development (Meinicke, 1916).

Growth models and yield tables have been developed to help guide forest management decisions through synthesizing the interaction between complex biological processes and silvicultural practices (Skovsgaard and Vanclay, 2008). But the roles of abiotic and biotic

* Corresponding author.
E-mail addresses: Alex.Woods@gov.bc.ca (A.J. Woods), martin_watts@telus.net (M. Watts).

https://doi.org/10.1016/j.foreco.2019.117558
Received 20 June 2019; Received in revised form 14 August 2019; Accepted 16 August 2019
Available online 03 September 2019
0378-1127/ Crown Copyright © 2019 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/BY-NC-ND/4.0/).

disturbance agents are often overlooked in these modelling exercises (Woods and Coates, 2013). Similarly, most dynamic global vegetation models used to forecast broad-level ecosystem change do not explicitly consider the effects of biotic disturbance agents but instead implicitly assume that the impacts of tree death are accounted for in the temporally and spatially averaged 'background' mortality rate (McDowell et al., 2015). The concept of 'background' mortality is an oversimplification used to compensate for a lack of knowledge of biotic disturbance agent caused damage in general (McDowell et al., 2011), and particularly so for forest pathogens. Apart from wildfires, outside of the fields of forest pathology and entomology, tree mortality is thought to be driven primarily by inter-tree competition (Peet and Christensen, 1987; Bazzaz, 1990; Lutz and Halpern, 2006). Pathogens have been ignored and or misunderstood by most forest managers and ecologists, especially in the case of native forest pathogens (Hansen and Goheen, 2000). In managed forests no single factor, not site index, species composition nor growth rates, has a greater influence on modelled stand productivity than unexpected loss of crop trees (Flewelling and Monserud, 2002). Trees do not die without cause (Waring, 1987; Franklin et al., 1987) but determining the actual cause and mechanism is not simple (Das et al., 2016). Accurate forest disease assessments can provide valuable insights into managed stand growth and should be standard practice, particularly at mid-rotation when the loss of intended crop trees can be so influential.

Root disease fungi are some of the most dominant forest disturbance agents, responsible for losses of forest carbon storage as great as those of wildfire and timber harvesting (Healey et al., 2016). The pathogens spread primarily by root contacts between infected and uninfected susceptible host roots (Hansen and Goheen, 2000) though spore dispersal is important for some root disease fungi (Pukkala et al., 2005). Root diseases are natural components of most forest ecosystems and have been considered agents of diversity, facilitating the creation of wildlife habitat among other ecological roles in unmanaged forests (van der Kamp, 1991). They exist at spatial and temporal scales far exceeding that of their hosts. A single clone of Armillaria root disease, for example, is estimated to occupy an area of close to 1000 ha and is between 1900 and 8700 years old (Ferguson et al., 2003). Given the spatial and temporal scale at which root disease pathogens can exist in relation to their hosts, large-scale, long-term monitoring plots are most appropriate for studying their behaviour and impacts in managed forests.

Tomentosus root disease (*Onnia tomentosa* (Fr.) P.Karst) causes root and butt rot of several spruce species as well as other tree species in boreal and sub-boreal forests across Canada (Lewis, 1997) and northern temperate forests of Eurasia (Hansen et al., 2018). The fungus is found throughout north-central and northwest British Columbia (Lewis and Hansen, 1991; Woods, 2003) and in these areas it is the dominant fungal pathogen of interior spruce (*Picea engelmanni* Parry ex Engelm. x *Picea glauca* (Moench) Voss), attacking lodgepole pine (*Pinus contorta* var. *latifolia* Dougl. ex Loud.) to a lesser extent. The pathogen causes a decline in tree vigour due to a loss of root function and structural weakening leading to standing mortality or windthrow (Whitney, 1993; Lewis, 1997). From a timber perspective, the greatest impacts of this pathogen may be due to butt cull that can exceed 30% of merchantable volume of infected trees (Lewis, 1997). Lewis and Hansen (1991) proposed that prompt re-establishment of susceptible hosts on tomentosus root disease sites would allow more time for initial stump root-tree root contacts to develop and, therefore, more time for the disease to spread, leading to increased losses.

In our study we assess the influence of a biotic disturbance agent, tomentosus root disease, over more than half of the planned rotation for a random sample of managed interior spruce stands. We examine the apparent spread from infected stumps of the previous stand to new host trees and then from tree to tree as the stands approach 50 years of age. Relatively little literature has addressed root disease development and tree to tree spread within mid-rotation managed stands, particularly in the case of tomentosus root disease. We then assess the growth and yield of individual crop trees and relate that growth to the presence of root disease across the sampled stands. To the best of our knowledge, our study is unique in terms of its combination of scale, both spatially and temporally, and in its sampling of managed stands without prior knowledge of the pathogen of interest. It is this combination of study attributes that allows us to critically examine the blind-spot that exists between traditional forest growth forecasts and the behaviour and impacts of an overlooked and underestimated biotic disturbance agent.

Our objectives were to: 1. Report changes in host tree health in relation to a biotic disturbance agent, tomentosus root disease, over a 16-year period in mid-rotation managed spruce plantations, 2. Estimate the volume losses (reduced growth and tree mortality) to tomentosus root disease and compare that loss to other causes of tree mortality (i.e., unexplained background mortality) to illustrate the importance of a specific pathogen, 3. Assess whether changes in health status for trees within a given proximity of infected stumps and neighbouring infected trees are consistent with accepted knowledge of root disease behaviour, 4. Illustrate how a biotic disturbance agent in mid-rotation managed stands can challenge traditional growth and yield concepts, such as stable crop tree density, with implications for the accuracy of productivity estimates and sustainable forest management.

2. Methods

2.1. Study area description

Our study was conducted in the coastal transition forests of the Kispiox Timber Supply Area (TSA) in northwest British Columbia (BC), Canada. Based on the biogeoclimatic ecological classification (BEC) system used in BC, the dominant biogeoclimatic zone in the study area is the Interior Cedar Hemlock (ICH) zone (Banner et al., 1993). Forests here contain a wide variety of tree species including western hemlock (*Tsuga heterophylla* (Rafn.) Sarg.), subalpine fir (*Abies lasiocarpa* (Hook.) Nutt., amabilis fir (*Abies amabilis* (Dougl. ex Loud.) Dougl. ex Forbes), western redcedar (*Thuja plicata* Donn ex D. Don), interior spruce, lodgepole pine, paper birch (*Betula papyrifera* Marsh.), trembling aspen (*Populus tremuloides* Michx.) and black cottonwood (*Populus balsamifera* L.).

2.2. Mid-rotation managed interior spruce stands

In 2000, sample plots were established in 10 randomly selected stands from the population of spruce-leading plantations (≥50% spruce) aged 20–30 years in the ICH zone of the Kispiox timber harvesting land-base (see Woods, 2003). The population contained just over 50 stands as clear-cut harvesting was not common practice in the area prior to the late 1960s. Our 10 stand sample should be representative of the oldest cohort of managed stands in the TSA. Stands were selected without prior knowledge of the presence or incidence of any biotic disturbance agent. Silviculture records were reviewed for each of the 10 stands to determine their forest management history. Eight of the 10 stands had been broadcast burned prior to planting while the remaining two were planted directly following harvest. All 10 stands were planted with interior spruce and we assumed that given the forest management era, the seedlings were two-year-old, bare-root stock. Based on the silviculture records, planting densities ranged from about 900 to 1400 with a mean of 1120 stems per hectare.

2.3. Plot establishment, tree measurements and root disease detection

In each selected stand a 100 × 100 m plot was randomly located by overlaying a 1:20,000 scale forest cover map with a 200 × 200 m grid, then randomly selecting one of the grid points to serve as the northwest corner of the plot. The height (to 0.1 m), diameter at breast height (DBH; measured at 1.3 m to 0.1 cm), root disease status and spatial



Fig. 1. Increment coring to detect tomentosus root disease (*Onnia tomentosa*) at the base of a ~40 year old interior spruce tree in the Itzul plot of the Kispiox TSA, sporocarps in foreground of inset photo, and an extracted tree core showing the characteristic colour and pattern of decay. Note orientation of core in photo is reversed, healthy wood is found just inside bark, also note pink stained drill chips to the left of the core. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

location for all trees > 1.5 m in height was then recorded. In total, data were collected for over 12,000 individually tagged trees. The species, root disease status and location of all stumps > 20 cm diameter at the root collar from the former stand were also stem-mapped at plot establishment. Stem-mapping was conducted using a Total Station survey instrument mounted atop an 8 m-tall scaffolding tower located near the center of the plots. Individual trees and stumps were spatially located by placing a survey reflecting prism mounted on an extendable height pole at their location.

The determination of root disease status for stumps was based on the presence of pitted decay, a pattern of wood decay characteristic of tomentosus root disease colonization that can be identified with confidence up to 40 years after tree death (Lewis and Hansen, 1991). In instances where we could not be certain of the disease status of stumps due to the state of decomposition, we recorded their infection status as unknown. All spruce trees were sampled with a cordless drill using a 10 mm diameter drill bit at or close to the root collar. The presence of red-brown stain and white pitted-decay was used to confirm tomentosus infection. Trees with inconclusive drill chips were cored using an increment core for more positive disease identification (Fig. 1). All disease diagnoses were based on visual identification of stain and decay patterns in infected wood.

In 2016 all previously tagged trees were re-measured for DBH and heights and were assessed for general health condition and form. Trees that had not previously been tagged due to their small size at the time of plot establishment were not included. All spruce trees were once again examined for the presence of pink to brick-red stain and advanced pitted decay by drilling into the base of the tree within 10 cm of the root collar using a 10 mm drill and where necessary an increment corer.

2.4. Monitoring changes in tomentosus root disease incidence over time

For each plot we compared the condition of individual trees at plot establishment to their condition 16 years later and grouped them based on any changes. We tallied trees that were uninfected in 2000 and 2016, (coded as B/B) separately from those that became infected but were still alive in 2016 (B/C), and from those that had died from other causes (B/D), as well as those that had recently died from tomentosus (B/E). Similarly, we identified trees that were live and infected in 2000 and 2016 (C/C), separately from those initially live infected trees that had died by 2016 (C/E). We also identified the trees that were dead and infected at the time of plot establishment (E/E). We compared the volume growth between assessments of all but the last of these root disease condition groups. There were no uninfected dead trees recorded at plot establishment. Current root disease impacts were estimated based on the volume of both live and dead infected trees. Tree volumes were calculated using measured DBH and height and the 2002 Kozak BEC-based taper function (Kozak, 2004) which predicts the stem diameter at 0.1 m height intervals, calculates the volume of each stem section using Smalian's formula then sums the sections to get total tree volume.

2.5. Changes in the spatial distribution of tomentosus root disease influence over time

To map and assess tomentosus root disease infection zones, we divided the 100 × 100 m plots into 0.1 × 0.1 m pixels. We examined circular infections zones around each infection source using radii from 0.5 to 5.0 m, by 0.5 m increments. Using the Pythagorean Theorem, we determined if each pixel was in an infection zone by calculating the distance from each infection source to the centroid of each pixel. If the distance was less than or equal to the infection zone radius, the pixel was in the infection zone. A database for each pixel recorded the number of infection zones it was in (disease intensity) subdivided by the infection source type (stump, live infected tree or dead infected tree). The pixel database contained two variants, 2000 and 2016, as infected trees could change from live to dead between assessments. Finally, we created maps using the R software ggplot2 package (Wickham, 2016) of each plot showing the location of potential infection zones differentiating between infected stumps, infected trees both live and dead at plot establishment and infected trees both live and dead at our second assessment in 2016. Maps were also created for the infection zone intensity. We chose a four meter potential infection zone radius for our maps based on the work of Lewis and Hansen (1991) who found the probability of infection transferring from 30-year old spruce stumps to regenerated spruce trees was 25% at two meters, 10% at 3.75 m and who did not find the fungus more than five meters from stumps.

2.6. Determination of site index (SI)

We determined site index, the height of the largest diameter trees at breast height age 50, for each of the 10 plots by finding the mean height of largest DBH spruce tree in 81 randomly located 0.01 ha circular plots. We did not determine tree age in the field but instead used total age based on the age since planting, assuming a planting stock age of two years. All site index determinations were made using the Goudie (1984) height growth equation for plantation white spruce via the Site Tools v4.1 software (BC Ministry of Forests, Lands and Natural Resource Operations, 2017).

2.7. Spruce tree growth compared to modelled expectations

We compared current standing volume of spruce and of all conifer species combined to growth and yield expectations by modelling each stand with the growth and yield model TASS II v2.7.75 (see reference Mitchell, 1975). We assumed our randomly located plots were representative of their respective stands which allowed us to compare per hectare model estimates to our measured values. We used the initial planting density and available interim silviculture survey information to adjust modelled stand densities up to the time of our initial plot establishment in 2000, and site index as determined from the 2016 plot measurements as model inputs (Table 1). We netted down model forecasts of volume using the standard TASS operational adjustment factors (OAFs). OAF 1 is a constant multiplier of 0.85 used to account for natural gaps in full stocking caused by a variety of biotic and abiotic factors and OAF 2 is a multiplier whose impact increases annually to account for decay and mortality due to biotic factors and is typically set at 0.95 at the index age of 100 (Di Lucca, 1999).

3. Results

3.1. Host composition and root disease condition of the former stands

Based on our assessment of stumps, interior spruce and lodgepole pine, the preferred host tree species for tomentosus root disease, were present in the former stands prior to harvest across all 10 sites (Table 2). The susceptible host component of the former stands varied from a low of 3% at Centre Nash Y to a high of 63% at Kispiox Trail. At the time of plot establishment we found evidence of tomentosus root disease in the interior spruce stumps in all 10 plots and in lodgepole pine stumps in nine of the 10 plots (Table 2). The incidence of tomentosus root disease in susceptible host species stumps varied from a low of 8% at Cancel to a high of 92% at Burdick. The number of infected stumps/ha, spruce and lodgepole pine combined, ranged from a low of two at Cancel to a high of 194 at 10-Link Olive. On average, the proportion of stumps infected was 64% for spruce and 62% for pine.

3.2. Species composition and changing health condition of spruce trees over a 16-year period mid-rotation

As the managed interior spruce stands approached 50 years of age, the initially planted species averaged just over 80% of total conifer stocking and ranged from 64% at Nangeese to 97% at Centre Nash Y (Table 3). The remainder of stocking consisted of natural ingress of a mixture of conifers and broadleaf species. Based on our assessment of stumps at the time of plot establishment, the decision to plant interior spruce following harvest increased host availability for tomentosus root disease by between 32 and 94% compared to the former species composition of each plot, respectively.

Tomentosus root disease was present in second growth spruce trees in all 10 of the randomly selected plots at the time of plot establishment. Across all plots combined, the incidence of tomentosus infected trees increased from 5% in 2000 (mean stand age 28) to 14% in 2016 (mean stand age 44) (Table 3). This single forest pathogen accounted for 97% of all unhealthy trees, with broken tops or toppled stems caused by heavy snow accounting for the remainder of compromised trees. At every plot we observed trees deemed healthy at our first assessment that had subsequently become infected and died over the 16-year period between assessments. Overall, 10% of trees that showed no sign of root disease infection in 2000 were infected by 2016 and of these 22% were dead. Of those 5% of spruce trees that had been identified as live infected in 2000, 29% were dead by 2016.

Over the 16-year period between assessments the range of tomentosus incidence increased from 0.4 – 9.4% in 2000 to 2.5–26.7% in 2016 but the prevalence of infected trees appeared unrelated to the species composition of the former stands (Fig. 2). The former stand with the highest spruce component, Kispiox Trail, had the third lowest current disease incidence while three of the top four stands in terms of current disease incidence, Nash Y, Cancel and Center Nash Y, had the three lowest pre-harvest spruce proportions.

We found considerable variability in root disease activity among plots with two, Burdick and Nash Y, suffering the greatest overall incidence of tomentosus root disease, with 23 and 27% of host spruce trees infected on each site, respectively. These two plots also experienced the greatest changes in incidence since the prior assessment with 18 and 19%, respectively, of trees previously considered healthy showing signs of infection or dying from the disease by 2016 (Fig. 2). It is worth noting that these two most heavily root disease infested stands differed in age by 7 years with the highest incidence stand, Nash Y, being one of the youngest stands and Burdick one of the oldest. These two stands also differed widely in terms of prior stand inoculum with Burdick having close to five times the number of infected stumps as that found at the Nash- Y site, 162 infected spruce and pine stumps vs 33 (Table 2).

The proportional breakdown by diameter class of the live infected trees in 2016 closely resembled that of healthy trees (Fig. 3). The increase in disease incidence between assessments was evenly distributed across diameter classes and not confined to the smallest trees. To date tomentosus root disease has tended to kill smaller trees but we observed trees over 25 cm DBH that had died from the disease. Over the past 16 years close to 30% of live infected trees died, so we expect larger diameter trees will continue to succumb to this disease. Even if the larger infected trees survive to rotation there will be significant losses to timber quality and merchantable volume from decay and butt cull, as

Table 1
Silviculture data used as TASS model inputs in 10 one-hectare plots located in mid-rotation spruce leading plantations in the Kispiox TSA of northwest BC.

	Nash Y	Burdick	Cancel	Center Nash Y	10-Link Olive	Tea Lakes	Itzul	Kispiox Trail	Kuldo	Nangeese
Planting year	1977	1970	1970	1977	1970	1969	1976	1970	1972	1970
Site index (m)	21.6	22.1	23.4	22.7	20.4	18.7	25.0	21.3	22.9	20.5
Initial planting density	1111	1214	1403	1197	916	881	1197	1119	1001	1119
Interim silv. survey year	1986	1988	1983	1986	1988	1990	1990	1987	1990	1987
Natural ingress spp*	Hw	Pl	Hw	Cw	Hw	Pl	Ep	Pl	Act	Hw
Natural ingress density	1859	364	406	802	1862	428	130	530	1458	2032
Density at plot establishment (2000)	1363	1382	1281	1449	1133	567	1210	1380	808	1764

* Hw = western hemlock, Pl = lodgepole pine, Cw = western redcedar, Ep = paper birch, Act = black cottonwood.

Table 2
Frequency of host species stumps and all stumps > 20 cm diameter at the root collar from the former stands in 10 one-hectare plots located in mid-rotation spruce leading plantations in the Kispiox TSA of northwest BC.

	Plot									
Former stand	10-Link Olive	Burdick	Cancel	Center Nash Y	Itzul	Kispiox Trail	Kuldo	Nangeese	Nash Y	Tea Lakes
Spruce stumps	137	106	13	6	68	163	68	37	28	100
Pine stumps	88	84	1	4	1	2	21	3	30	34
Infected spruce stumps	123	98	1	4	49	85	26	14	21	47
Infected pine stumps	71	64	1	2	1	0	3	2	12	11
All Stumps	598	467	261	361	269	261	376	487	350	526

Table 3
Stand age and counts of both total trees and interior spruce (host) trees with the latter further differentiated on the basis of tomentosus root disease condition and changes in that condition between two assessments over a 16 year period in 10 one-hectare stem-mapped plots located in mid-rotation interior spruce plantations in the Kispiox TSA.

	Nash Y	Burdick	Cancel	Center Nash Y	10-Link Olive	Tea Lakes	Itzul	Kispiox Trail	Kuldo	Nangeese	Mean
Age in 2000	23	30	30	23	30	31	24	30	28	30	29.0
Condition* in 2000/2016											**Total**
B/B	767	762	658	1160	777	340	986	1209	696	1100	8455
B/C	148	125	84	134	63	35	49	35	17	15	705
B/D	3	7	7	5	1	0	6	12	3	4	48
B/E	37	46	23	27	14	1	18	18	1	8	193
C/C	59	36	62	39	33	17	40	26	5	4	321
C/E	27	24	12	29	13	1	16	5	4	1	132
E/E	9	1	6	9	14	1	7	4	3	0	54
All spruce	**1050**	**1001**	**852**	**1403**	**915**	**395**	**1122**	**1309**	**729**	**1132**	**9908**
% spruce	77	73	67	97	81	69	93	95	92	64	80
All trees	**1371**	**1364**	**1289**	**1452**	**1132**	**573**	**1209**	**1376**	**795**	**1761**	**12,322**

* B - Live and not infected, C - Live and Infected, D - Dead and not infected, E - Dead and Infected.

evidenced by our disease detection methods (see Fig. 1).

Across the 10 plots combined just over 91% of all interior spruce volume remained healthy but considerable variability existed among plots, with a low of just under 80% at Nash Y to a high of just over 98% at Nangeese (Table 4). Overall, the volume of interior spruce trees that was healthy at plot establishment and remained so 16 years later (B/B) grew from an area weighted mean of 39.0 to 140.6 m^3/ha. Trees with evidence of root disease infection at the time of our second assessment had reduced volume growth compared to trees that had remained disease free. The extent of growth reduction was dependant on a combination of infection duration and severity. The volume growth of trees that were initially healthy but became live infected (B/C) was 17% less across all plots combined than the growth of trees that remained healthy, while the volume growth of trees that were live infected since plot establishment (C/C) was 28% less than that of trees that remained healthy (Table 4). The trees that were initially healthy but died by 2016 (B/E) clearly experienced a greater reduction in growth than any of the trees that remained alive (B/C and C/C) and the trees that were live infected at plot establishment but subsequently died had the greatest overall loss of growth (B/C < C/C < B/E < C/E). Except for the Nash Y plot, volume growth reduction of infected trees followed this categorical ranking. At Nash Y, trees that were healthy at plot establishment and became live infected by 2016 (B/C) grew 39% less in volume than trees that remained healthy. But these more recently

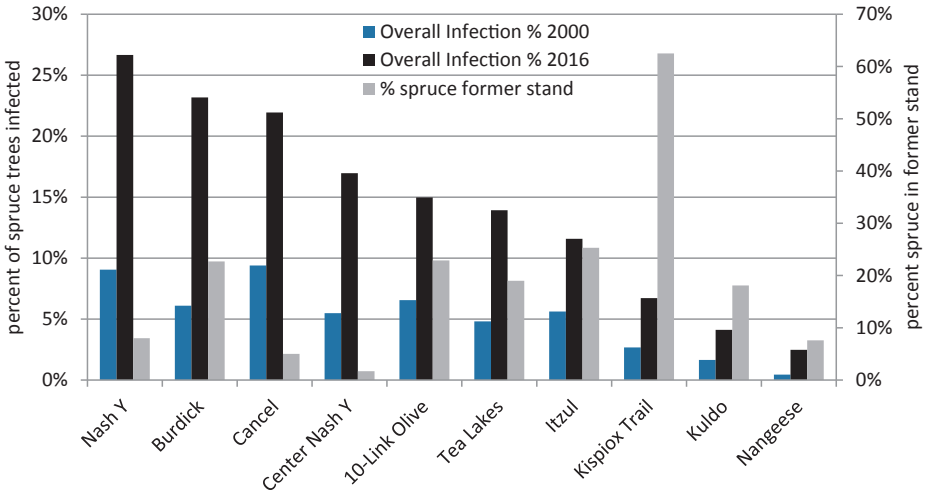


Fig. 2. Incidence of tomentosus root disease in interior spruce stands aged 23–31 at the time of plot establishment in 2000 and 16 years later (left axis), sorted in order of highest incidence in 2016, and the percent spruce composition of the former stand (right axis) in 10 one-hectare stem-mapped plots in the Kispiox TSA of northwest BC.

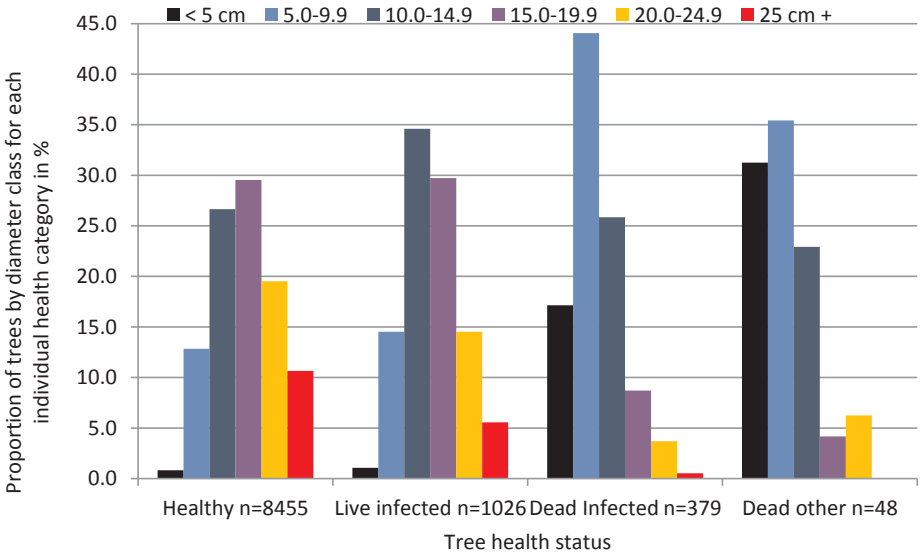


Fig. 3. Proportion of interior spruce trees by diameter class in 2016 in percent in each of four health condition categories; healthy, live tomentosus infected, dead tomentosus infected and dead other causes across 10 one-hectare plots located in mid-rotation spruce stands in the Kispiox TSA of northwest BC. (Note differences in tree counts between health categories).

infected trees grew even less than the trees that were live infected at the time of plot establishment (C/C) which grew 24% less than the healthy trees (Table 4).

We also found considerable variability in both the volume of healthy trees and the growth of that volume among plots. The Nash Y plot had the lowest volume of healthy spruce trees at plot establishment and the second lowest volume of healthy spruce at the time of the 2016 assessment, 83.4 m^3 (Table 4). Curiously, healthy trees at Nash Y had the second greatest overall volume growth at 369%, behind only Itzul, the most productive plot, where healthy trees grew from 42.6 m^3 at plot establishment to 218.2 m^3 16 years later, an increase of 412%. Importantly, the Nash Y and Itzul stands differ in age by only one year, differ in spruce stocking by just over 70 sph or about 7% (Table 3), differ in site index by 3.4 m, (Table 1) yet differ in live spruce volume by 131 m^3/ha and in healthy spruce volume by 135 m^3/ha or by more than 160% (Table 4). This volume difference is far greater than expected if site index was the single dominant factor differentiating the performance of these two stands. This finding emphasizes the uncertainty associated with forecasting managed stand yields without regard for biotic disturbance agents like tomentosus root disease.

3.3. Changes in the spatial distribution of tomentosus root disease over time

We tried to better understand the distribution of infected trees at our second assessment by separating the potential inoculum sources; stumps from the prior stand and infected trees at plot establishment. The location of infected trees at stand age ~30 was not always closely associated with the location of infected stumps from the prior stand (Fig. 4a). More surprisingly, the location of the recently infected trees, those that were healthy at the time of plot establishment and had subsequently become infected, were not necessarily closely associated with the location of infected trees at the time of plot establishment (Fig. 4b). In plots like 10-Link Olive, as the stand approached age 50 the location of the most recently infected trees (shown as red and blue triangles) was best captured by a combination of both the infected stumps from the prior stand and the infected trees at about age 30 (Fig. 4c).

But the pattern of disease distribution and spread that we found in the group of the plots consisting of 10-Link Olive, Itzul, Burdick and Kuldo (Fig. 5a, b, c and f) and Tea Lakes (Supplemental Fig. S1) was not repeated in the Nangeese, Kispiox Trail and Centre Nash Y plots (Supplemental Figs. S2–S4) nor in Cancel and Nash Y (Fig. 5d and e). In the

Table 4
Interior spruce volume (m^3/ha) at plot establishment and 16 years later for the cohort of trees that were: initially healthy and have remained so (B/B), initially healthy then infected and alive (B/C), initially healthy then died from other causes (B/D), initially healthy then died from tomentosus (B/E), initially infected and have remained alive (C/C), initially infected and have died from tomentosus (C/E) and those trees that were dead and infected by tomentosus at plot establishment (E/E). The area weighted mean values (right column) are weighted by stand area (as a fraction of the total area sampled) to better reflect the population average.

	Nash Y	Burdick	Cancel	Center Nash Y	10-Link Olive	Tea Lakes	Itzul	Kispiox Trail	Kuldo	Nangeese	Mean (area wtd).
Stand area (ha)	24.1	44.5	34	17.5	97.2	22	11.4	80	32.8	43.8	
B/B 2000 (m^3/ha)	17.75	34.77	57.42	44.67	38.11	22.93	42.64	43.45	36.60	40.83	38.96
B/C 2000 (m^3/ha)	4.63	6.28	6.39	3.98	2.53	1.76	2.27	1.23	0.85	0.58	2.80
B/D 2000 (m^3/ha)	0.02	0.16	0.26	0.03	0.01	0.00	0.02	0.39	0.06	0.05	0.13
B/E 2000 (m^3/ha)	0.49	1.84	1.05	0.52	0.40	0.13	0.49	0.14	0.10	0.16	0.51
C/C 2000 (m^3/ha)	1.16	1.32	4.29	1.32	0.66	0.80	1.38	0.97	0.16	0.23	1.09
C/E 2000 (m^3/ha)	0.42	0.50	0.46	0.90	0.20	0.13	0.56	0.04	0.05	0.06	0.25
E/E 2000 (m^3/ha)	0.05	0.00	0.12	0.04	0.25	0.01	0.05	0.02	0.02	0.00	0.08
Total 2000 (m^3/ha)	24.47	44.87	69.87	51.43	41.90	25.75	47.37	46.21	37.81	41.91	43.82
B/B 2016 (m^3/ha)	83.43	131.24	186.09	162.74	125.59	68.09	218.18	157.10	168.10	136.28	140.60
B/C 2016 (m^3/ha)	15.06	20.54	18.29	13.12	7.81	4.84	10.08	3.82	3.48	1.71	8.85
B/D 2016 (m^3/ha)	0.04	0.19	0.49	0.06	0.02	0.00	0.04	1.00	0.02	0.13	0.28
B/E 2016 (m^3/ha)	1.10	2.99	2.18	1.11	0.68	0.24	1.24	0.20	0.19	0.21	0.91
C/C 2016 (m^3/ha)	4.41	3.82	10.95	4.08	1.73	1.87	5.44	3.01	0.68	0.52	3.13
C/E 2016 (m^3/ha)	0.60	0.63	0.73	1.28	0.24	0.23	1.21	0.48	0.06	0.08	0.34
Total 2016 (m^3/ha)	104.63	159.41	218.73	182.38	136.05	75.27	236.18	165.17	172.52	138.91	154.20

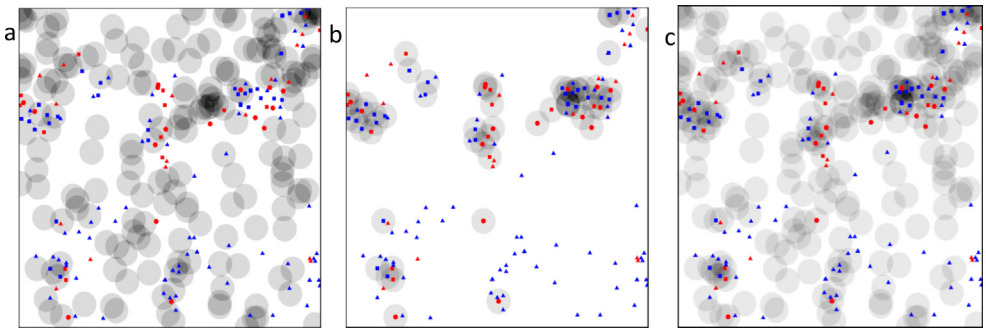


Fig. 4. Three representations of the stem mapped location of tomentosus root disease at the 10-Link Olive one hectare plot, where 4 m radius potential infection zones shown as grey disks are based on; (a) the location of infected stumps from the prior stand, (b) the location of infected trees at plot establishment (~age 30), and (c) the combination of both stumps and infected trees at plot establishment. Red circular dots represent dead infected trees in 2000 (E/E), squares represent trees that were alive and infected in 2000; blue being those remaining alive in 2016 (C/C) and red being those that died (C/E). Triangles represent trees that were disease free in 2000 and became infected by 2016, blue being live (B/C) and red the dead (B/E). Healthy trees are not shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

latter group the location of infected trees in 2016 appeared to have little association with either the location of infected stumps from the prior stand or the location of infected trees 16 years earlier.

In general we found the incidence and distribution of tomentosus root disease in both stumps and trees varied widely among plots. The 10-Link Olive plot (Figs. 4 and 5a) contained almost 200 infected spruce and pine stumps yet as the stand approached age 50, had a lower incidence of infection in trees, less evenly distributed across the plot, than the Cancel plot (Fig. 5d). Cancel had a relatively even distribution and the third highest overall incidence of infected trees despite containing

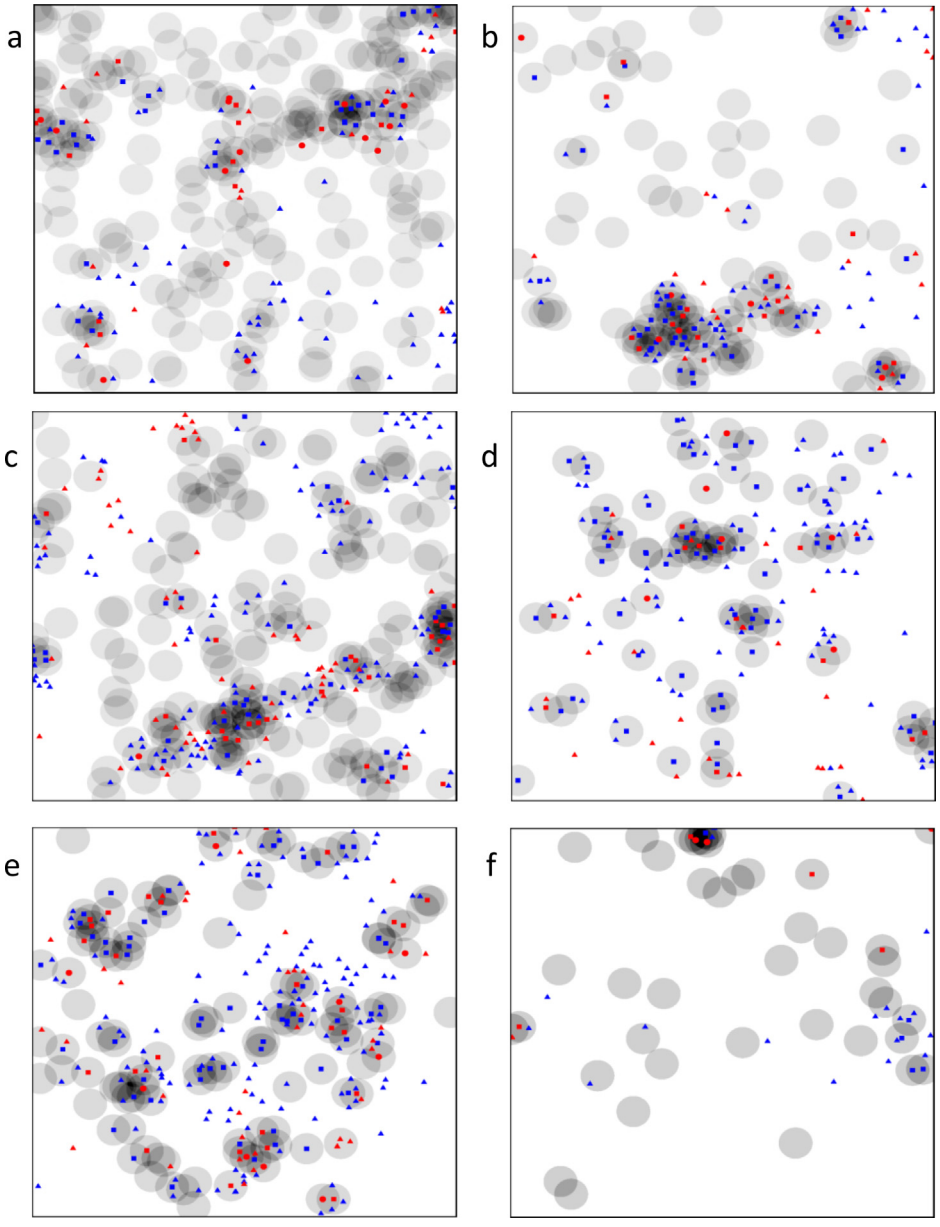


Fig. 5. Differences in the incidence and the distribution of tomentosus root disease infected trees and stumps in six (chosen to show the range of variability) of the 10 one-hectare stem-mapped mid-rotation interior spruce plantations (a. 10-Link Olive, b. Itzul, c. Burdick, d. Cancel, e. Nash Y, f. Kuldo). Red dots represent dead infected trees in 2000 (E/E) while squares represent trees that were live and infected in 2000; blue being those remained alive in 2016 (C/C) and red being those that died (C/E). Triangles represent trees that were disease free in 2000 and became infected by 2016, blue being live (B/C) and red the dead (B/E). Grey disks represent 4 m potential infection influence radii, those with squares in the centre being live infected trees in 2000, those with red circles being dead infected in 2000 and those without anything in the centre being infected stumps, darker shades represent increasing overlap of potential infection radii. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

only two infected stumps from the former stand. Stands with the highest root disease incidence, Burdick (Fig. 5c) and Nash Y (Fig. 5e) differed markedly in the number of infected stumps, the former containing close to five times that of the latter (Table 2). These two highest incidence plots also differed considerably in the spatial distribution of infected trees at the time of our second assessment. Burdick having a more uneven distribution of infected trees than Nash Y. Lower incidence plots such as Itzul and Kuldo (Fig. 5b and f) contained the same number of spruce stumps from the prior stand but the former had twice the incidence of infection in those stumps and more than twice the incidence of infection in spruce trees by the time of the second assessment. Across all 10 plots, over the 16 years between assessments the increase in the extent of plot area occupied by tomentosus infected trees was considerable (more triangles than squares) but the complexity of the relationships made any generalizations about root disease behaviour difficult to make.

3.4. Measured plot volumes compared to modelled mid-rotation expectations

Using a combination of silviculture records to initialize TASS, and our initial plot condition in 2000 as an interim model adjustment point, we found model projections of spruce volume exceeded measured plot volume at our second assessment in 2016 by a weighted average of over 30% after accounting for standard assumed losses with default OAFs (Table 5). The difference between modelled and measured spruce volumes varied widely among plots from a low of 6% after assumed losses at Itzul to 43% at Nash Y. In the next three most heavily infested plots, Center Nash Y, Cancel, and Burdick, modelled spruce volumes exceeded measured total spruce volumes by 15, 25, and 29%, respectively, again after standard assumed rates of loss were applied to model forecasts (Table 5). But root disease was not the only agent responsible for the differences between measured plot volumes and modelled projections. At Nangeese, the lowest root disease incidence plot, there was still a discrepancy in spruce volume of over 39% after standard loss factors were applied (Table 5). Much of the discrepancy between measured and modelled spruce volume at this plot appeared to be due to natural ingress of other conifer species displacing the planted species. At mid-rotation, natural ingress of other conifer species was helping to offset the loss of spruce reducing the difference between measured and modelled total volumes across several of the plots. There remained, however, a close to 17% difference in the weighted mean between measured and modelled volume for all species combined after losses were accounted for with default OAFs.

4. Discussion and conclusions

We examined the influence of a biotic disturbance agent, tomentosus root disease, on the oldest managed stands in a previously unmanaged landscape allowing us to witness pathogen expression in a first rotation setting over the longest duration available. Our assessments, first at close to stand age 30 when mortality caused by this disease usually begins (Whitney, 1993) and when the signs of disease are still clearly identifiable in stumps from the former stand (Lewis and Hansen, 1991) and the second 16 years later, show how disease expression changes over time at an infrequently and insufficiently monitored stage of stand development. Our large stem-mapped plots capture the spatial variability of root disease expression and their size, coupled with their random selection without any prior knowledge of the pathogen, makes our study unique. Our examination of mid-rotation managed stands illustrates the degree to which a lesser known native forest pathogen can interfere with traditional forest management goals and objectives. The unique combination of study attributes allows us to critically examine the blind-spot that exists between traditional forest management growth and yield forecasts and the behaviour and impacts of an overlooked and underestimated biotic disturbance agent.

We did not target known damaged stands, and although a single pathogen was dominant, if our methods were replicated elsewhere we suspect a different biotic disturbance agent or combination of disturbance agents could well lead to similar results. In an extensive assessment of managed stands across the forested landscape to the east of our study, Woods et al. (2017) found a wide variety of disturbance agents affecting managed stand growth and productivity at levels beyond that expected in growth forecasts (i.e., OAFs). Those authors suggested climate change as a possible factor in the unexpectedly high rates of damage and loss. Based on over 100 years of monitoring, some of the most intensively studied managed forests, those of central Europe, are reported to have experienced increased growth rates due to climate change (Pretzsch et al., 2014), but this study explicitly excluded plots damaged by storms or bark beetle infestations. Such reductionist approaches to monitoring forest dynamics, where disturbance damaged plots are removed from data sets, are the norm in traditional forestry (Bruce, 1977). The removal of disturbance damaged trees and plots from studies may improve the precision of growth estimates for undamaged trees but such actions risk missing the larger picture. The prediction of the future states of trees, stands and landscapes can no longer be made with precision (Millar et al., 2007). The uncertainty surrounding future managed stand productivity revealed by the unexpectedly wide variation in mid-rotation stand growth driven by an overlooked biotic disturbance agent is one of the key findings of our study.

From a forest management perspective our 10 mid-rotation spruce plantations would be assumed to follow similar growth trajectories aimed at achieving comparable timber volumes at rotation. They were all 20–30 years old at plot establishment and were all located in the ICH biogeoclimatic zone, one of the most complex and productive ecological zone in BC (Coates et al., 1997), with site indices ranging from a low of 19 m to a high of 25 m. Eight of the 10 stands had been broadcast burned for site preparation prior to planting. All but two of the stands were planted at densities between 1000 and 1200 sph, the exceptions

Table 5
Comparison of 2016 measured plot total volumes to TASS II v2.07.75 modelled total spruce and combined conifer volumes, including standard assumed net-down Operational Adjustment Factors (OAFs) designed to account for losses to biotic and abiotic factors, for 10 stem mapped plots located in mid-rotation spruce stands of the Kispiox TSA.

	Nash Y	Burdick	Cancel	Center Nash Y	10-Link Olive	Tea Lakes	Itzul	Kispiox Trail	Kuldo	Nangeese	Mean (area wt)*
Plot vol all spp (m^3)	157.50	230.17	276.60	187.54	175.95	105.57	249.18	180.27	185.19	221.06	194.37
Plot vol spruce (m^3)	104.63	159.41	218.73	182.38	136.05	75.28	236.18	165.17	172.52	138.91	154.12
Plot vol healthy spruce (m^3)	83.43	131.24	186.09	162.74	125.59	68.09	218.18	157.10	168.09	136.28	140.60
TASS vol all spp after OAFs (m^3)	188.00	277.10	293.30	215.50	216.60	125.60	252.30	253.40	215.50	235.20	233.10
TASS vol spruce after OAFs (m^3)	183.80	226.00	292.80	215.40	208.60	93.50	251.10	248.10	214.10	229.50	221.78
% diff. all spp vol. after OAFs	16.23	16.94	5.69	12.97	18.77	15.95	1.24	28.86	14.06	6.01	16.62
% diff. spruce volume after OAFs	43.07	29.46	25.30	15.33	34.78	19.49	5.94	33.43	19.42	39.47	30.51
% diff. of healthy spruce after OAFs	54.61	41.93	36.44	24.45	39.79	27.18	13.11	36.68	21.49	40.62	36.60

* Area weighted based on stand area as a fraction of the total stand area sampled.

being Tea Lakes at 900 sph and Cancel at 1400 sph. Yet despite these similarities, the current conditions in these stands varied widely and unexpectedly. For instance the oldest plantation, Tea Lakes, had the lowest mid-rotation volume of spruce at 75 m^3/ha while one of the youngest plantations, Itzul, had almost three times that spruce volume at 236 m^3/ha, a difference far greater than that explained by site index alone. We believe the wide variation among plots is due to a complex combination of factors with tomentosus root disease playing a central but not exclusive role. Although we know the initial planting densities for each of the plantations, our plots were not installed at stand establishment so the number of planted spruce trees that died from root disease or any other cause prior to stand age 30 is unknown. Similarly, although we have interim silviculture surveys for the plantations in which our plots are located, the timing of natural ingress initiation and possible losses to inter-tree competition is not precisely known. Our results clearly illustrate the degree of uncertainty associated with forecasting managed stand yields in the absence of more frequent monitoring updates of stand conditions.

Infected stumps from the previous stand are typically considered the primary source of root disease inoculum, responsible for spread of the pathogen to the following generation of trees (Vasaitis et al., 2008). Hunt and Peet (1997) estimated tomentosus root disease spread at an average rate of 20 cm/year in a combination of both mature and second growth spruce stands in central BC. In many instances we found new infections much further away than what would be expected based on their estimated spread rate (16 × 20 cm/year or 3.2 m). Lewis and Hansen (1991) found viable tomentosus root disease inoculum in 30-year old spruce stumps in north-central BC and found that spruce trees regenerated within two metres of infected stumps had a 25% chance of becoming infected while at four metres the probability of infection dropped to 10%. Our large stem-mapped plots revealed a more complex, less predictable relationship between root disease inoculum in 30-year old stumps and host trees in mid-rotation stands. In some plots root disease behaviour appeared to conform to expected norms of spread from either infected stumps or earlier identified infected trees in the current stand. In others, we found newly infected trees, including some recently killed by the disease, much further than the maximum reported distance for fungal spread from inoculum sources of 4.5 m (Lewis, 1990). While examining the influence of tomentosus root disease on differences in the dynamics of unmanaged and partial-cut forests of east-central BC, Newbery et al. (2007) found that the relationship between inoculum levels and subsequent disease spread was inconsistent. The same authors suggested part of the explanation for their findings could be due to a larger role of spores in the spread of this disease. Gibson (2005) found that the population genetic structure of *Onnia tomentosa* (=*Inonotus tomentosus*) was typical of a fungus that spreads by sexual reproduction (basidiospores) with local and limited asexual spread by root contacts. Our study took place in the same location and over the same time period as an unprecedented Dothistroma needle blight epidemic (*Dothistroma septosporum* (Dorog.) Morelet.) (Woods et al., 2005). Bohaychuk and Whitney (1973) found *I. tomentosus* spore discharge was greatest during periods of high relative humidity (> 85%) and moderate temperatures of 16–22 °C, conditions similar to those known to favour Dothistroma (Woods et al., 2016). It may be more than coincidence that an apparent increase in spore mediated root disease spread took place during a period of rising temperatures and increased summer precipitation which would make this already complex host/pathogen relationship even less predictable.

On average the measured total spruce volume in our sample stands was 30.5% less than that projected by TASS. Tomentosus root disease was largely responsible for the much lower productivity we observed. Our measured volume closely matched model forecasts on the Itzul plot, the plot with the highest site index and one of the highest spruce species compositions. If all of our plots had been located on comparable high productivity sites, had maintained similar spruce stocking with low rates of loss to root disease and other disturbances, model forecasts may have more closely aligned with measured volumes across our studied stands. The TASS model accurately forecasts individual tree volumes in controlled research settings (Goudie, 1998) but may lose accuracy when applied on a larger operational scale as a result of natural gaps in stocking caused by a variety of factors including pathogens and other disturbance agents. A lack of stochasticity often revealed when modelling efforts developed from small plots are applied to forests (Pukkala et al., 2005) may be part of the explanation. More fundamentally, the TASS model is based on the same foundational tenets that are the dominant paradigm in forestry where the prediction of site productivity is based on the assessment of dominant stand height, Eichhorn's rule (which implies that when the dominant tree height is known, stand yield can be easily estimated now and then projected into the future) and the thinning response hypothesis (Skovsgaard and Vanclay, 2008). Few forest growth models include the effects of disturbance events which may limit the reliability of the model results (Kirilenko and Sedjo, 2007). Our results reflect the influential role of a biotic disturbance agent that is operating on a different scale and following a different set of rules.

Given the significant increase in incidence and spatial distribution of infected trees, their reduced rate of growth and increased rate of mortality over the 16 year period we observed, not to mention butt cull, and that final harvest is still 30 years into the future, we expect the trend of decreasing spruce dominance to continue to rotation. As a result, we also expect the gap between model forecasts and realised volume to continue to widen. Butt cull alone can account for losses of more than 30% of merchantable volume for live infected spruce trees (Lewis, 1997). The former stands on which our plots were established contained lower proportions of susceptible hosts than the spruce dominated plantations yet had infection rates of over 60% within the susceptible species. In subsequent work we intend to forecast the continued spread of the pathogen and model the timber volume impacts at the projected harvest age using the spatially explicit data from our two assessments. Our results clearly show the importance of recognizing the role of specific disturbances agents and how those agents can compromise forest management objectives. Our results also show the critical need for long-term, large-scale monitoring of managed stand performance with direct feedback mechanisms capable of both challenging long held assumptions and informing current practices to avoid repeating mistakes of the past.

Past forest management efforts in our naturally species diverse study area focused on maximizing timber productivity of two conifer species, lodgepole pine and interior spruce (Woods, 2003). Simplified management choices have been shown to lead to less resilient, more disease susceptible forests (Mildrexler et al., 2019) while promoting species diversity has been shown to increase stand resistance to natural disturbances (Jactel et al., 2017). Even in the most rapidly growing and diverse forests, mortality can occur much faster than associated recovery by growth to similar tree sizes and forest structure (Franklin et al., 1987; Frank et al., 2015). We found that a single host-specific biotic disturbance, tomentosus root disease, accounted for close to 90% of both the number and the volume of spruce trees that died over a 16-year period in stands aged 30–45. These losses were not due to 'background mortality' (see McDowell et al., 2011) that might accrue more or less equally to any species, but instead were species specific losses of the chosen managed species. The final impact of the disease will depend on a variety of factors including the length of time that infected trees survive, their rate of growth until death, the rate of decomposition of dead standing trees, and the extent of butt cull associated with live infected trees at harvest. These losses are far greater than those that would have occurred had the inherent species diversity of this forested landscape been maintained. Our results support the recommendations of Pawson et al. (2013) who called for moving away from centuries old traditional forestry practice of stand focused science to an approach that considers the landscape and the complex interactions among organisms, such as pathogens (Holdenrieder et al., 2004) that exist at that

larger scale. Given the increasingly complex relationships between pathogen and host in our study, we believe attempting to accurately model the system would be an overly ambitious venture. We recommend instead that forecasts of managed stand growth incorporate more caution, recognizing the inherent uncertainty associated with biotic and abiotic disturbances and their effects on forest growth and recommend that managers seriously consider the wider implications of overly optimistic projections.

Acknowledgements

We thank all those responsible for the initial plot establishment in 2000 including Hubert Burger, James Lucas, Larry McCulloch, Tim and Josie Qureshi, Bruce La Haie, Theresa White, Paul Hanna and others. We thank the Rainmakers Unit Crew from the BC Wildfire Service and others for clearing the accumulated vegetation from access routes to our plots after 16 years of growth. We also thank Kathy Fraser and Adrian de Groot and others for their diligence in data collection at our second plot assessment in 2016 and Ken Polsson of the Forest Analysis and Inventory Branch of BC Ministry of FLNRORD who completed our TASS model runs. Erin Holtzman helped with our data management, figure development and volume growth comparisons and we thank her for her assistance. We also thank Dave Coates, Hardy Griesbauer, Stefan Zeglen, Catherine Bealle Statland, Peter Ott, Jevan Hanchard, Paul Hennon and three anonymous reviewers for their reviews of earlier drafts of this work. Funding was provided by the BC Ministry of Forests, Lands and Natural Resource Operations, Forest Health program.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2019.117558.

References

Banner, A., MacKenzie, W., Haeussler, S., Thompson, S., Pojar, J., Trowbridge, R., 1993. A field guide to site identification and interpretation for the Prince Rupert Forest Region. Research Branch, BC Ministry of Forests, 31 Bastion Square, Victoria BC.

Bazzaz, F.A., 1990. Plant – plant interactions in successional environments. In: Grace, J.B., Tilman, D. (Eds.), Perspectives on Plant Competition. Academic Press, San Diego, CA, pp. 239–263.

BC Ministry of Forests Lands and Natural Resource Operations, 2017. Site Tools v4.1 for white spruce.

Bohaychuk, W.P., Whitney, R.D., 1973. Environmental factors influencing basidiospore discharge in *Polyporus tomentosus*. Can. J. Bot. 51, 801–815.

Bruce, D., 1977. Yield differences between research plots and managed forests. J. For. 1, 14–17.

Coates, K.D., Banner, A., Steventon, D., LePage, P., Bartemucci, P., 1997. The Date Creek Silvicultural systems study in the Interior Cedar-Hemlock forests of northwestern British Columbia: overview and treatment summaries. Land Manage. Handb. 38, B.C. Min. For., Res. Branch, Victoria, B.C.

Das, A.J., Stephenson, N.L., Davis, K.P., 2016. Why do trees die? Characterizing the drivers of background tree mortality. Ecology 97 (10), 2616–2627.

Di Lucca, C.M., 1999. TASS/SYLVER/TIPSY: Systems for predicting the impact of silvicultural practices on yield, lumber value, economic return and other benefits. In: Bamsey, C.R. (Ed.), Stand Density Management Conference: Using the Planning Tools. Clear Lake Ltd., Edmonton, AB, Canada, pp. 7–16.

Ferguson, B.A., Dreisbach, T.A., Parks, C.G., Filip, G.M., Schmitt, C.L., 2003. Coarse-scale population structure of pathogenic Armillaria species in a mixed-conifer forest in the Blue Mountains of northeast Oregon. Can. J. For. Res. 33, 612–623.

Flewelling, J.W., Monserud, R.A., 2002. Comparing methods for modelling tree mortality. In: Crookston, N.L., Havis, R.N. (Eds.), Second Forest Vegetation Simulator Conference. USDA, Forest Service, Rocky Mountain Research Station. Proc. RMRS-P-25, Ogden, UT, pp. 168–177.

Flower, C.E., Gonzalez-Meler, M.A., 2015. Response of temperate forest productivity to insect and pathogen disturbances. Annu. Rev. Plant Biol. 66, 547–569.

Franklin, J.F., Shugart, H.H., Harmon, M.E., 1987. Tree death and as ecological process. Bioscience 37 (8), 550–556.

Frank, D., Reichstein, M., Bahn, M., Thonicke, K., Frank, D., Mahecha, M.D., Smith, P., Van der Veld, M., Vicca, S., Babst, F., Beer, C., Buchmann, N., Canadell, J.G., Ciais, P., Cramer, W., Ibrom, A., Miglietta, F., Poulter, B., Rammig, A., Senevirante, S.I., Walz, A., Wattenbach, M., Zavala, M.A., Zscheischler, J., 2015. Effects of climate extremes on the terrestrial carbon cycle: concepts, processes and potential future impacts. Glob. Change Biol. 2015 (21), 2861–2880.

Gibson, S., 2005. An examination of genetic variation and disease spread of the fungal pathogen, *Inonotus tomentosus* (MSc. Thesis). University of Northern British Columbia, Prince George, BC.

Goudie, J.W., 1984. Height growth and site index curves for lodgepole pine and white spruce and interim managed stand yield tables for lodgepole pine in British Columbia. B.B. Min. For., Res. Br. Unpubl. Rep. p. 75.

Goudie, J.W., 1998. Model validation: A search for the magic grove or magic model. In: Bamsey, C.R. (Ed.), Stand density management: Planning and implementation. Pp. 45–58. Edmonton, AB.

Hansen, E.M., Lewis, K.J., Chastagner, G.A., 2018. Compendium of Conifer Diseases, second ed. American Phytopathological Society, St Paul MN. USA, pp. 224.

Hansen, E.M., Goheen, E.M., 2000. *Phellinus weirii* and other native root pathogens as determinants of forest structure and process in western North America. Annu. Rev. Phytopathol. 38, 515–539.

Healey, S.P., Raymond, C.L., Lockman, B., Hernandez, A.J., Garrard, C., Huang, C., 2016. Root disease can rival fire and harvest in reducing forest carbon storage. Ecosphere 7 (11), 16. https://doi.org/10.1002/ecs2.1569.

Holdenrieder, O., Pautasso, M., Weisberg, P.J., Lonsdale, D., 2004. Tree diseases and landscape processes: the challenge of landscape pathology. Trends Ecol. Evol. 19 (8), 446–452.

Holling, C.S., Meffe, G.K., 1996. Command and control and the pathology of natural resource management. Cons. Biol. 10 (2), 328–337.

Hunt, R.S., Peet, F.G., 1997. Annual spread rate of Tomentosus root disease. Plant Dis. 81 (9), 1053–1056.

Jactel, H., Bauhus, J., Boberg, J., Bonal, D., Castagneyrol, B., Gardiner, B., Gonzalez-Olabarria, J.R., Koricheva, J., Meurisse, N., Brockerhoff, E.G., 2017. Tree diversity drives forest stand resilience to natural disturbances. Curr. Forest. Rep. https://doi.org/10.1007/s40725-017-0064-1.

Kirilenko, A.P., Sedjo, R.A., 2007. Climate change impacts on forestry. Proc. Nat. Acad. Sci. 104, 19697–19702.

Kosiba, A.M., Meigs, G.W., Duncan, J.A., Pontius, J.A., Keeton, W.S., Tait, E.R., 2018. Spatiotemporal patterns of forest damage and disturbance in the northeastern United States: 2000–2016. For. Ecol. Manage. 430, 94–104.

Kozak, A., 2004. My last words on taper equations. Forest. Chronicle 80 (4), 507–515.

Lewis, K.J., 1990. Epidemiology and Population Biology of *Inonotus tomentosus* as a Pathogen of Forests of British Columbia. PhD. thesis. Oregon State University.

Lewis, K.J., 1997. Growth reduction in spruce infected by *Inonotus tomentosus* in central British Columbia. Can J. For. Res. 27, 1669–1674.

Lewis, K.J., Hansen, E.M., 1991. Survival of *Inonotus tomentosus* in stumps and subsequent infection of young stands in north central British Columbia. Can. J. For. Res. 21, 1049–1057.

Lutz, J.A., Halpern, C.B., 2006. Tree mortality during early forest development: a long-term study of rates, causes and consequences. Ecol. Monog. 76, 257–275.

McDowell, N.G., Beerling, D.J., Breshears, D.D., Fisher, R.A., Raffa, K.F., Stitt, M., 2011. The interdependence of mechanisms underlying climate-driven vegetation mortality. Trends Ecol. Evol. 26 (10), 523–532.

McDowell, N.G., Coops, N.C., Beck, P.S., Chambers, J.Q., Gangodagamage, C., Hicke, J.A., Huang, C.Y., Kennedy, R., Krofcheck, D.J., Litvak, M., Meddens, A.J., 2015. Global satellite monitoring of climate-induced vegetation disturbances. Trends Plant Sci. 20 (2), 114–123.

McNamee, P., Sutherland, G., Shaw III, C.G., Stage, A., Byler, J., Crookston, N., et al., 1989. Description of a multi-species root disease model developed for silvicultural planning and management in coniferous forests of western North America. In: Morrison, D.J. (Ed.), Proceedings of the Seventh IUFRO Conference on Root and Butt rot in forest trees, August 1988. Forestry Canada, Pacific Forestry Centre, Victoria, BC, pp. 320–325.

Meinicke, E.P., 1916. Forest Pathology in Forest Regulation. USDA Bull. No. 275. P. 62.

Mildrexler, D.J., Shaw, D.C., Cohen, W.B., 2019. Short-term climate trends and the Swiss needle cast epidemic in Oregon's public and private coastal forestlands. For. Ecol. Manage. 432 (2019), 501–513.

Millar, C.I., Stephenson, N.L., Stephens, S.L., 2007. Climate change and forests of the future: managing in the face of uncertainty. Ecol. Appl. 17, 2145–2151.

Mitchell, K.J., 1975. Dynamics and simulated yield of Douglas-fir. For. Sci. Monogr. 17, 39.

Newbery, J.E., Lewis, K.J., Walters, M.B., 2007. Inonotus tomentosus and the dynamics of unmanaged and partial-cut wet sub-boreal spruce-fir forests. Can J. For. Res. 37, 2663–2676.

Pawson, S.M., Brin, A., Brockerhoff, E.G., Lamb, D., Payn, T.W., Paquette, A., Parrotta, J.A., 2013. Plantation forests, climate change and biodiversity. Biodivers. Conserv. 2013 (22), 1203–1227.

Peet, F.G., Hunt, R.S., 2005. A new simulator for the spread of forest root diseases by individual root contacts. Forest Sci. 51 (5), 425–437.

Peet, R.K., Christensen, N.L., 1987. Competition and tree death. Bioscience 37 (8), 586–595.

Pretzsch, H., Biber, P., Schütze, G., Uhl, E., Rötzer, T., 2014. Forest stand growth dynamics in Central Europe have accelerated since 1870. Nat. Commun. 5, 4967.

Puettmann, K., Coates, K.D., Messier, C., 2009. A Critique of Silviculture: Managing for Complexity. Island Press, Washington DC.

Pukkala, T., Möykkynen, T., Thor, M., Rönnberg, J., Stenlid, J., 2005. Modelling infection and spread of *Heterobasidion annosum* in even-aged Fennoscandian conifer stands. Can J. For. Res. 35, 74–84.

Seidl, R., Rammer, W., Jager, D., Lexer, M.J., 2008. Impact of bark beetle (*Ips typographus* L.,) disturbance on timber production and carbon sequestration in different management strategies under climate change. For. Ecol. Manage. 256 (2008), 209–220.

Skovsgaard, J.P., Vanclay, J.K., 2008. Forest site productivity: a review of the evolution of dendrometric concepts for even-aged stands. Forestry 81, 13–31.

Vasaitis, R., Stenlid, J., Thomsen, I.M., Barklund, P., Dahlberg, A., 2008. Stump removal

to control root rot in forest stands. A literature study. Silva Fennica 42 (3), 457–483.
Van der Kamp, B.J., 1991. Pathogens as agents of diversity in forested landscapes. Forest. Chronicle 67 (4), 353–354.
Waring, R.H., 1987. Characteristics of trees predisposed to die. Bioscience 37, 569–574.
Whitney, R.D., 1993. Damage by Tomentosus root rot in white spruce plantations in Ontario, and the effects of thinning on the disease. For. Chron. 69 (4), 445–449.
Wickham, H., 2016. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag, New York, pp. 2016.
Woods, A.J., 2003. Species diversity and forest health in northwest British Columbia. Forest. Chronicle 9 (5), 892–897.
Woods, A., Coates, K.D., Hamann, A., 2005. Is an unprecedented Dothistroma needle blight epidemic related to climate change? Bioscience 55, 761–769.
Woods, A.J., Coates, K.D., 2013. Are biotic disturbance agents challenging basic tenets of growth and yield and sustainable forest management? Forestry 86, 543–554.
Woods, A.J., Marti-Garcia, J., Bulman, L., Vasconcelos, M.W., Boberg, J., La Porta, N., Peredo, H., Vergara, G., Ahumada, R., Brown, A., Diez, J.J., 2016. Dothistroma needle blight, weather, and possible climate triggers for the disease's recent emergence. Forest Pathol. 46, 443–452.
Woods, A.J., Coates, K.D., Watts, M., Foord, V., Holtzman, E.I., 2017. Warning signals of adverse interactions between climate change and native stressors in British Columbia forests. Forests 8 (8), 280. https://doi.org/10.3390/f8080280.